



Republic of the Philippines
Department of Education
National Capital Region
Schools Division Office of Las Piñas City

THREE- TERM DEFINITIVE BUDGET OF WORK (DBOW)
REVISED K TO 10 CURRICULUM
S.Y. 2026-2027
SCIENCE 9

FIRST TERM	
CONTENT STANDARD	• Newton's laws explain and predict how objects move due to the forces that act on them. • Electricity is a flow of electrons and can be measured and understood using current, voltage, and resistance in circuits. • Electromagnetic radiation travels using transverse waves of different wavelengths. • Scientists and engineers use electromagnetic radiation to design modern technologies that benefit people and society. • Evidence for continents moving includes jig-saw matching of coastlines, rock types, and the presence of similar fossils in places separated by vast distance. • The movement of lithospheric plates provides a theory for understanding Earth's geological history.
PERFORMANCE STANDARD	By the end of the term, learners demonstrate a practical understanding of Newton's three laws of motion to describe relationships between variables and use these to explain everyday application of Newton's laws. Through practical investigations, learners demonstrate qualitative understanding of the features of electricity and apply their understanding of electrical circuitry in homes. Learners exhibit skills in gathering information from secondary sources to describe the frequencies across the electromagnetic spectrum and identify practical applications and detrimental effects that electromagnetic radiation may have on living things. Also, learners exhibit skills in evaluating information

	from secondary sources, and draw on their scientific understanding of the location and geological setting of the Philippines to explain its unique landforms and dynamic geologic activity in a global context.
SUGGESTED ACTIVITIES	• Inertia at Home Investigation – Learners test inertia using common objects (coin-on-card over a glass, book-and-paper pull, toy car on a smooth surface), then describe what changed the motion and what stayed the same. • Acceleration and Net Force Push-Pull Lab – Learners use improvised carts (toy car, bottle on cardboard, or small box) and apply different pushes/pulls to observe changes in speed and direction; learners record observations using a simple motion table. • Force–Mass–Acceleration Relationship Ramp Investigation – Learners roll the same object down a ramp with different masses (adding coins/clay) or apply the same pull with different masses (rubber band pull) to observe how mass affects acceleration; record and compare results. • Action–Reaction Pairs Community Observation Walk – Learners identify action–reaction pairs in familiar Philippine settings (stepping off a bangka, pushing a tricycle, jumping on the ground, book on a table) and create force diagrams for each scenario. • Force Diagram Clinic Using Real Situations – Learners practice drawing force diagrams for objects at rest and in motion (hanging bag, jeepney slowing down, bicycle moving, box being pushed), focusing only on forces that act on the object. • EMR as Transverse Waves Rope-and-Slinky Modeling – Learners use a rope or slinky to model transverse waves (crest, trough, wavelength, frequency) and connect the model to how electromagnetic radiation travels as transverse waves. • EM Spectrum Ordering with Frequency–Wavelength Strip Activity – Learners arrange EM wave cards (radio to gamma) and use a simple ruler-scale strip to compare relative wavelength and frequency; include local examples (cellphone signal, Wi-Fi, microwave oven, TV remote IR, sunlight/UV). • EMR Applications in Philippine Life Research Mini-Task – Learners gather secondary-source information on EMR applications common in the Philippines (telecommunications, radio/TV, medical x-ray, UV disinfection) and summarize how each benefits people.

	● Harmful Effects of EMR Evidence Brief – Learners use secondary sources to explain harmful effects of EMR (UV and skin/eye damage, excessive exposure risks, improper x-ray exposure), linking effects to human activities and safety practices in daily life. ● Continental Drift Evidence Jigsaw Puzzle Lab – Learners do a “jigsaw” matching activity using coastline pieces, rock-type strips, and fossil distribution cards to explain why continents were once joined; include Philippine/Asian region reference maps where available. ● Philippine Plate Boundary Evidence Map Task – Learners examine PH-focused maps (trenches, faults, volcano belts, earthquake distribution) and describe what topographical and geological patterns suggest about plate boundaries near the Philippines. ● Plate Boundary Types Model Lab – Learners model divergent, convergent, and transform boundaries using cardboard/foam/paper strips and record the structures/events produced (ridges, trenches, mountains, faults, earthquakes).
SUGGESTED PERFORMANCE TASK	● Action–Reaction Rescue Vehicle Model – Design and build a model vehicle that is propelled by action–reaction forces and requires an understanding of how forces influence the motion of the vehicle. ● Philippine Electricity Generation and Distribution System Model – Design and build a working model, schematic diagram, or a full poster representing the electricity generation and distribution system in the Philippines. ● Electromagnetic Spectrum Media Poster – Design and build a media/technology poster depicting the uses, benefits, and harmful effects of different electromagnetic waves.

CONTENT	WEEK NO.	LEARNING COMPETENCIES AND SPECIFIC OBJECTIVES	NUMBER OF DAYS TAUGHT
Newton's Laws, Force, and Energy	Week 1-2	Identify inertia as the tendency for an object to stay at rest or in motion unless acted on by an unbalanced net force.	2
		Day 1: Identify inertia as the tendency of an object to stay at rest or in motion unless acted upon by an unbalanced net force.	
		Day 2: Recognize real-life situations that demonstrate inertia in objects at rest or in motion	
		Demonstrate in practical situations and describe that acceleration is a change in speed and/or direction as the result of a net force.	2
		Day 3: Identify the factors that describe acceleration: change in speed (magnitude), change in direction, and the time interval over which the change occurs.	
		Day 4: Demonstrate acceleration by manipulating forces on an object in a classroom setting to change its speed and/or direction.	
		Investigate the relationship among force, acceleration, and mass.	2
		Day 5: Investigate how changes in force affect the acceleration of an object when mass is kept constant.	
		Day 6: Analyze the relationship between mass and acceleration when the applied force remains the same.	
		Explain that when any two objects interact, there are equal but opposite forces exerted between them, which is evident in many practical situations and applications.	2
		Day 7: Identify action and reaction force pairs in various everyday situations.	
		Day 8: Explain Newton's Third Law through various hands-on activities.	
		Observe and identify action-reaction pairs in everyday situations such as stepping off a boat, or a book on a table, and draw force diagrams to explain how the pairs affect the motion of objects.	2
		Day 9: Observe and identify action-reaction pairs in simulated everyday situations.	

		Day 10: Draw force diagrams to explain how action-reaction pairs affect the motion of objects.	
Electric Current, Electrical Circuits, Interpreting Patterns in Data, and Electromagnetic Waves	Week 3	Identify that electricity is a flow of electrons and show appreciation for the need to observe safe measures in handling electricity.	2
		Day 11: Describe electricity as a flow of electrons.	
		Day 12: Demonstrate appreciation for the importance of observing safety measures when handling electricity.	
		Participate in guided investigations to infer the relationship among current, voltage, and resistance in assembled series and parallel circuits with varying number of loads and battery.	5
		Day 13: Identify and define the three fundamental quantities of electricity—current, voltage, and resistance — and their corresponding units of measurement (Amperes, Volts, and Ohms)	
		Day 14: Recognize and label the standard schematic symbols for a battery, bulb, connecting wire, switch, and resistor in a provided circuit diagram.	
		Day 15: Explain the functional difference between a series and a parallel circuit by describing the continuous path of electron flow in each configuration.	
		DAY 16 (Day 1-15): SUMMATIVE TEST	1
Electric Current, Electrical Circuits, Interpreting Patterns in Data, and Electromagnetic Waves	Week 4-8	Participate in guided investigations to infer the relationship among current, voltage, and resistance in assembled series and parallel circuits with varying number of loads and battery. (Continuation)	
		Day 17: Predict the effect on the remaining loads (bulbs) when one load is removed or disconnected from both a series and a parallel circuit.	
		Day 18: Compare and contrast the behavior of voltage and current in series versus parallel circuits using a Venn diagram or a comparison table.	
		Draw diagrams of and assemble series and parallel circuits, showing switch, battery, loads/resistors, ammeter, and voltmeter.	6
		Day 19: Describe the difference between series and parallel circuits.	
		Day 20: Explain the function of an ammeter and voltmeter in a circuit.	

		Day 21: Draw a simple series circuit showing a switch, battery, resistor/load, ammeter, and voltmeter.	
		Day 22: Draw a simple parallel circuit showing a switch, battery, resistor/load, ammeter, and voltmeter.	
		Day 23: Construct working circuit of series and parallel using basic electrical components.	
		Day 24: Construct working circuit of series and parallel using basic electrical components.	
		Collaborate in a class discussion to recognize the advantages and limitations of using series or parallel circuits.	4
		Day 25: Identify the basic characteristics of series and parallel circuits.	
		Day 26: Recognize the advantages and limitations of series circuits.	
		Day 27: Recognize the advantages and limitations of series circuits.	
		Day 28: Recognize the appropriate use of series and parallel circuits in practical applications.	
		Describe electromagnetic radiation (EMR) as energy that is created by the vibrations of electrically charged particles which allows it to travel through materials or space as transverse waves.	3
		Day 29: Describe electromagnetic radiation.	
		Day 30: Explain electromagnetic radiation as transverse waves that travel through material or space.	
		Day 31: Trace the oscillation of energy in the material or space as transverse waves.	
		DAY 32: (Day 17-31): SUMMATIVE TEST	1
		Compare the relative wavelengths and frequencies of different types of electromagnetic waves, including radio waves, microwaves, infrared, visible light, ultra-violet, x-rays, and gamma radiation.	5
		Day 33: Describe relative wavelengths and frequencies of EM waves.	
		Day 34: Explain the relationship between relative wavelength and frequency.	
		Day 35: Distinguish the different types of electromagnetic waves.	

		Day 36: Arrange electromagnetic wave cards (radio to gamma) and use a simple ruler-scale strip to compare relative wavelength and frequency	
		Day 37: Compare relative wavelengths and frequencies using the EM spectrum	
		Identify practical applications of electromagnetic radiation, such as radio waves used in telecommunications, and x-rays and gamma rays in medicine.	2
		Day 38: Cite the uses of electromagnetic radiation such as radio waves in telecommunication.	
		Day 39: Recognize the uses of electromagnetic radiation such as x-rays and gamma rays in medicine.	
		Gather information from secondary sources to explain the harmful effects that EMR can have on living things.	3
		Day 40: Gather information from secondary sources about the harmful effects of electromagnetic radiation on living things.	
		Day 41: Identify and organize the harmful effects of electromagnetic radiation based on information from secondary sources.	
		Day 42: Explain the harmful effects of electromagnetic radiation on living things	
Plate Boundaries	Week 9-11	Identify and explain evidence that current continents are separate parts of what was a single continent millions of years ago.	5
		Day 43: Recognize how the Continental Drift Theory was developed	
		Day 44: Determine the fossils of plants and animals as evidence of the drifted continents	
		Day 45: Describe Pangaea and explain its difference to the existing continents of the earth at present	
		Day 46: Describe seafloor spreading process	
		Day 47: Determine the rate of seafloor spreading using magnetic polarity map	
		Participate in a collaborative group or class task to examine and describe the topographical and geological evidence for plate boundaries occurring in the area where the Philippines is located.	5
		Day 48: Identify plate boundaries around the Philippines using a map	

		Day 49: Analyze topographical features around the plate boundaries of the Philippines such as Trenches and Volcanoes.	
		Day 50: Examine geological evidence of plate boundaries around the Philippines.	
		Day 51-52: Create a group presentation on the evidence of plate boundaries around the Philippines.	
		Describe the types of plate boundaries found around the Earth.	3
		Day 53: Differentiate the types of plate boundaries (divergent, convergent, and transform) based on their structures.	
		Day 54: Explain how plate motions result in the formation of diverse tectonic boundaries.	
		Day 55: Compare and differentiate the characteristics and effects of each type of plate boundary.	

SECOND TERM

CONTENT STANDARD	• The geological time scale organizes major stages in the history of the Earth over more than 4 billion years. • Radioactive decay of material inside the Earth since it was formed is its internal source of energy. • The Earth’s interior is made up of layers of varying characteristics. • Models represent the size, structure, and relationship of components of the Solar System. • Observable evidence and models help explain the nature and origin of the Solar System. • High biodiversity means populations are more likely to overcome adverse conditions. • Human activities can adversely affect animals and plants in a variety of ecosystems.
PERFORMANCE STANDARD	By the end of the term, learners demonstrate an appreciation of the size and scale of the Earth and describe evidence for a dynamic Earth over its long geological history as well as the evidence that is used to build a model for the internal structure of the Earth. Learners demonstrate curiosity and open-mindedness in extending their knowledge and understanding of the dynamic Earth to evaluate evidence for theories for the formation of the Solar System. They describe modern scientific processes and technologies that are used by scientists to investigate the nature and evolution of the Solar System and Universe. They also explain that high levels of diversity help to maintain stability of an ecosystem. They research to classify critically endangered plants and animals of the Philippines and to identify strategies to protect and conserve them. They use drawings and diagrams to describe features of typical Philippine ecosystems and they conduct a survey to explore possibilities to minimize the impact of human activities.
SUGGESTED ACTIVITIES	• Philippine Biodiversity Classification Task – Learners use secondary sources to classify selected Philippine species as critically endangered, endangered, or vulnerable and identify threats affecting each. • Biodiversity Stability Modeling Activity – Learners model how higher biodiversity can increase ecosystem stability using a simple “food web balance” card model under

	stress scenarios (disease, food shortage, climate events). • Local Conservation Action Plan Mini-Research – Learners research how to protect and conserve endangered and/or economically important species in their locality and write a practical, community-level action list. • Fossil Dating Simulation Using Stratigraphy Cards – Learners use “layer cards” with fossils to infer relative ages of rocks and explain how fossils help date sediments and rock layers. • Relative vs Absolute Dating Classroom Investigation – Learners compare two mock datasets: stratigraphic layers (relative dating) and simplified radioactive decay results (absolute dating) to explain how geologic time subdivisions are determined. • Geologic Time Scale Timeline Build – Learners construct a scaled classroom timeline (string or paper strip) to represent major divisions of geologic time and place key events (fossils, major transitions) to show how the time scale recounts Earth’s history. • Seismic Wave Evidence Lab (S- and P-wave Data) – Learners analyze simplified seismogram/travel-time data to infer Earth’s internal layers and explain how seismic waves helped develop the Earth structure model. • Earth Layers Scale Drawing Activity – Learners create a scale drawing showing relative thicknesses of Earth’s layers (crust, lithosphere, asthenosphere, mantle, outer core, inner core) using provided thickness values and a chosen scale.
SUGGESTED PERFORMANCE TASK	• Space Technologies and Data Collection Report – Produce a report on space technologies (telescopes, probes) and explain their use in collecting data that contributes to our understanding of the Solar System. • Environmental Action Group Report – Write a report on an environmental action group and their impact on the biosphere and environments. • Philippine Ecosystem Biodiversity Research Project – Conduct a research project on a specific Philippine ecosystem and explain the advantages of high biodiversity in maintaining the stability of that ecosystem. • 3D Earth Features Model – Design and build a 3D model of the features of the Earth.

CONTENT	WEEK NO.	LEARNING COMPETENCIES AND SPECIFIC OBJECTIVES	NUMBER OF DAYS TAUGHT
Scale, Proportion, and Quantity, Structure of the Earth, and Geologic Time	Week 1-4	Describe how fossils can be used for dating the age of rocks and sediments.	3
		Day 1: Define fossils and identify the primary characteristics that make a species useful for dating geological strata.	
		Day 2: Explain how the Law of Superposition allows geologists to sequence rock layers from oldest to youngest based on their vertical position.	
		Day 3: Use a provided diagram of a sedimentary rock sequence to identify the relative ages of various fossils located at different layers.	
		Describe how relative and absolute dating techniques are used to determine the subdivisions of geologic time.	3
		Day 4: Define relative dating and absolute dating and give examples of each	
		Day 5: Describe how relative dating shows the order of events in geologic time	
		Day 6: Arrange the subdivision of geologic time in their proper chronological order using data from both dating techniques	
		Explain how the geologic time scale helps to recount the history of the Earth.	4
		Day 7: Identify the major divisions of the geologic time scale	
		Day 8: Describe important events that happened during different geologic eras	
		Day 9: Explain how the geologic time scale helps scientists recount earth's history using evidence from rocks and fossils.	
		Day 10: Construct a simple timeline showing the sequence of major geologic events.	
		Describe how seismic wave data has been used to develop a model for the internal structure and composition of the Earth.	5
		Day 11: Identify the two main types of seismic waves (body waves and surface waves)	
		Day 12: Distinguish P-waves and S-waves based on their properties	
		Day 13: Explain how seismic waves change speed and direction when passing through different materials inside the Earth	
		Day 14: Relate the behavior of seismic waves to the internal structure and composition of the Earth	

		Day 15: Describe how scientists used seismic data to create and improve the model of Earth's internal structure: crust, mantle and core	
		DAY 16 (Day 1-15): SUMMATIVE TEST	
		Create a scale drawing to represent relative thicknesses of the layers of Earth's interior, including the crust, lithosphere, asthenosphere, mantle, outer core, and inner core.	5
		Day 17: Identify the different Layers of the Earth.	
		Day 18: Describe the composition of the layers of the Earth.	
		Day 19: Differentiate the layers of the Earth according to their composition.	
		Day 20: List some natural phenomena happens in the Philippines related to the movement of the Earth's Surface.	
		Day 21: Illustrate through Drawing the following structure of the Earth layer according to their thickness.	
Origin of the Solar System	Week 5-6	Distinguish among comets, meteoroids, asteroids, and dwarf planets, and describe how they help us to understand the nature and formation of the Earth and the Solar System.	5
		Day 22: Describe the different celestial bodies in the Solar System.	
		Day 23: Enumerate theories on the formation of the Earth and the Solar System.	
		Day 24: Describe the relationships of celestial bodies to the formation of the Earth and Solar System such as Frost Line Theory and Earth's Water Delivery	
		Day 25: Describe the relationships of celestial bodies to the formation of the Earth and Solar System such as Frost Line Theory and Earth's Water Delivery	
		Day 26: Evaluate how analyzing the physical properties of meteorites allows scientists to reconstruct the chronological timeline and environment condition of the Earth.	
		Gather information from secondary sources to discuss the regular occurrence of meteor showers.	5
		Day 27: Describe meteor shower events.	
		Day 28: Identify the different secondary sources of meteor showers.	
		Day 29: Recognize some meteor shower events occurred in the Philippines.	
		Day 30: List some current events of meteor showers in Asia for the year 2025-2026 based on the data gathered from the secondary sources.	

		Day 31: Summarize the events of meteor showers among Asian countries based on secondary sources of information.	
		DAY 32: (Day 17-31): SUMMATIVE TEST	1
Space Technologies	Week 7	Explain how modern research about celestial objects uses new space technologies including telescopes and space probes.	5
		Day 33: Differentiate the different types of telescopes used in astronomical research according in their functions and capabilities.	
		Day 34: Explain how modern telescopes help scientists observe and gather information about distant celestial objects.	
		Day 35: Describe the functions and importance of space probes in exploring planets and other celestial bodies.	
		Day 36: Explain how advancements in space technologies contribute to a deeper understanding of the universe and modern astronomical discoveries.	
		Day 37: Identify different celestial objects studied by astronomers, such as stars, planets, moons, comets, asteroids, galaxies, and nebulae.	
Biodiversity and Endangered Species	Week 8-10	Explain the advantage of high biodiversity in maintaining the stability of an ecosystem during difficult conditions, such as food shortages, disease, and climate change.	3
		Day 38: Define Biodiversity and ecosystem; identify the living organisms that composes it.	
		Day 39: Compare ecosystems with high biodiversity and low biodiversity in relation to the food chain and environmental balance.	
		Day 40: Explain the advantage of high biodiversity in maintaining the stability of an ecosystem during difficult conditions, such as food shortages, disease, and climate change.	
		Use information from secondary resources to classify animals and plants of the Philippines as critically endangered, endangered, or vulnerable species.	4
		Day 41: Identify critically endangered, endangered, or vulnerable species in the Philippines.	
		Day 42: Describe critically endangered, endangered, or vulnerable species using secondary resources.	

		Day 43: Determine whether animals and plants of the Philippines are critically endangered, endangered, or vulnerable species using secondary resources.	
		Day 44: Classify animals and plants of the Philippines as critically endangered, endangered, or vulnerable species using secondary resources.	
		Discuss as a class how threats to biodiversity can lead to species extinction.	3
		Day 45: Identify threats to biodiversity that lead to species extinction (habitat loss, invasive species, pollution, population growth, overexploitation).	
		Day 46: Explain how these threats affect the extinction of species.	
		Day 47: Discuss ways to mitigate threats to diversity leading to species extinction.	
		Use information from secondary sources to research how to protect and conserve endangered and/or economically important species in the local community.	2
		Day 48: Identify endangered and economically important species in the local community using information from secondary sources.	
		Day 49: Explain ways of protecting and conserving endangered and economically important species based on information gathered from secondary sources.	
Types of Ecosystems in the Philippines	Week 10-11	Describe using labeled diagrams the biotic and abiotic features of tropical rainforests, swamps, estuaries, mangrove forests, and coral reefs.	2
		Day 50: Identify the biotic and abiotic components found in ecosystems such as tropical rainforests, swamps, estuaries, mangrove forests, and coral reefs.	
		Day 51: Describe the biotic and abiotic features of tropical rainforests, swamps, estuaries, mangrove forests, and coral reefs using simple labeled diagrams.	
		Use information from secondary sources to describe the possible effects of human activities, such as deforestation, pollution, and introduction of invasive species, on living things in an ecosystem	2
		Day 52: Identify the different human activities that affect an ecosystem using information from secondary sources	
		Day 53: Describe how these human activities affect living things in the ecosystem.	
		Plan to conduct a survey to explore the possibilities for minimizing the negative impacts of human activities on an ecosystem.	3
		Day 54: Identify different human activities that negatively affect ecosystems.	

		Day 55: Explain how human activities can negatively impact the ecosystem.	
		Day 56: Plan to conduct a survey to explore the possibilities for minimizing the negative impacts of human activities on an ecosystem.	

THIRD TERM

CONTENT STANDARD	• Transmission of traits is determined by DNA, genes, and chromosomes. • High biodiversity means populations are more likely to overcome adverse conditions. • Human activities can adversely affect animals and plants in a variety of ecosystems. • Valid and reliable scientific investigations include identification and control of variables. • Formation or breaking down of ionic or covalent bonds results in a chemical change. • Bonds are formed between atoms either by sharing or transferring of electrons. • The type of bond formed determines whether the result is a covalent or ionic compound. • Symbols for the elements are used as a basis for writing chemical formula of compounds. • The properties of pure substances depend on the type of bonding within them
PERFORMANCE STANDARD	By the end of the term, learners describe that the transmission of traits is determined by DNA, genes, and chromosomes. Learners also carry out a valid and reliable scientific investigation showing the formation of a new substance. They demonstrate an understanding of the significance of the valence electron of an element in the formation of bonds and identify bonds as ionic, covalent, or metallic. They use their knowledge of the symbols of elements to write the formula for a number of common compounds. They draw models of possible structures of ionic compounds and research the properties of ionic, covalent, and metallic substances. They use cartoons/comic strips to create interesting learning tools.
SUGGESTED ACTIVITIES	• DNA Double Helix Model Build – Learners construct a DNA model using low-cost materials (paper strips, beads, clay) and label key parts (sugar-phosphate backbone, base pairs) with a guided template. • Genes–Chromosomes–Traits Simulation – Learners use a simple trait-inheritance card model to explain how DNA, genes, and chromosomes work together to transmit traits. • Mutation Causes in Daily Life Evidence Task – Learners classify mutation causes (radiation, chemicals, infectious agents) using Philippine-relevant examples (UV exposure, smoking, certain chemicals) and summarize how each can change DNA/chromosomes. • Mutation Effects Case Card Analysis – Learners use secondary sources to categorize mutation effects (beneficial, harmful, neutral) using short case cards and explain why each falls in the category.

	• New Substance Formation Investigation (Precipitate or Carbonate Test) – Learners carry out a guided investigation showing formation of a new substance (e.g., CO_2 turning limewater milky; precipitate formation using teacher-provided safe solutions), then describe observed evidence of chemical change. • Bond Breaking and Bond Forming Model Activity – Learners use particle/bond models (paper clips, toothpicks, beads) to show that chemical change involves breaking existing bonds and forming new bonds. • Valence Electrons and Periodic Table Reasoning Task – Learners use periodic table position to identify valence electrons (focused example: oxygen) and explain why valence electrons matter in bonding. • Ion Formation and Ionic Bond Modeling – Learners model electron loss/gain using simple tokens to show ion formation and ionic bonding, using sodium chloride as the core example • Common Ionic Compounds Formula-and-Name Practice – Learners practice writing formulas and names for NaCl, MgO, KCl, and MgCl_2 using an ion-charge guide, then check using a peer key. • Covalent Bond Formation Model (H_2O and CO_2) – Learners model sharing of electrons to form covalent bonds using water and carbon dioxide, then write correct formulas and names for H_2O, CO_2, and NH_3. • Crystalline vs Molecular Structure Model Comparison – Learners build a “lattice” model for ionic compounds and a “separate molecules” model for covalent compounds to show structural differences. • Metallic Bonding ‘Sea of Electrons’ Demonstration Model – Learners use a simple model to represent metal cations in a sea of electrons and connect this to conductivity and malleability. • Properties of Substances Investigation Lab – Learners test and compare ionic, covalent, and metallic substances (teacher-provided) for hardness, conductivity (simple tester), and thermal response, then record patterns linking properties to bonding types.
SUGGESTED PERFORMANCE TASK	• Chemical Bonds Comic Strip – Create a cartoon/comic strip describing ionic, covalent, and metallic bonds and how they are involved in the formation of compounds.

CONTENT	WEEK NO.	LEARNING COMPETENCIES AND SPECIFIC OBJECTIVES	NUMBER OF DAYS TAUGHT
DNA Replication and Mutations	Week 1-3	Use models and labeled diagrams to represent the double helix structure of DNA (deoxyribonucleic acid).	4
		Day 1: Identify the main parts of the DNA double helix such as sugar, phosphate & nitrogen bases.	
		Day 2: Describe the double helix structure of DNA	
		Day 3: Label the parts of a DNA diagram correctly using a model or a picture	
		Day 4: Construct a DNA model or diagram	
		Explain the role of DNA, genes, and chromosomes in the transmission of traits.	4
		Day 5: Describe the role of DNA in the transmission of traits	
		Day 6: Explain how genes influence traits	
		Day 7: Describe how chromosomes function in carrying genes	
		Day 8: Discuss the role of DNA, genes, and chromosomes in the transmission of traits.	
		Describe mutations as changes in DNA or chromosomes and discuss some of the factors that cause mutations, such as infectious agents, radiation, and chemicals.	3
		Day 9: Define mutations as changes in DNA or chromosomes.	
		Day 10: Identify the factors that cause genetic mutations.	
		Day 11: Classify the factors (mutagens) that cause genetic mutations into three main groups: physical (radiation), chemical, and biological (infectious agents).	
		Use information from secondary sources to explain the beneficial, harmful, and neutral effects of mutations.	4
		Day 12: Identify beneficial, harmful, and neutral effects of mutations using secondary sources.	
		Day 13: Describe the beneficial and harmful effect of mutation to living things using secondary sources.	
		Day 14: Describe the natural effects of mutations using secondary sources.	
		Day 15: Discuss how beneficial, harmful, and neutral effects of mutations from information using secondary sources.	
		DAY 16 (Day 1-15): SUMMATIVE TEST	1

Valid and Reliable Investigations: Chemical Bonding	Week 4-6	Carry out a valid and reliable scientific investigation to show the formation of a new substance, such as formation of a carbonate (carbon dioxide in limewater), or formation of a precipitate (from silver nitrate solution).	5
		Day 17: Identify the parts of a scientific investigation.	
		Day 18: Describe evidence that indicates formation of a new substance.	
		Day 19: Explain the importance of controlling variables to ensure a scientific investigation is valid and reliable	
		Day 20: Carry out a scientific investigation to demonstrate the formation of a new substance, such as carbonate formation and precipitate formation.	
		Day 21: Apply proper laboratory safety protocols and systematic data recording during investigations involving chemical reactions.	
		Explain that the formation of new bonds or the breaking of existing bonds constitutes a chemical change and the formation of a new substance.	5
		Day 22: Identify observable evidence that indicates a chemical change has occurred.	
		Day 23: Describe how the breaking of existing bonds and formation of new bonds produce new substances.	
		Day 24: Explain that chemical change involves the breaking of existing bonds and the formation of new bonds resulting in new substances.	
		Day 25: Explain how bond breaking and bond formation lead to the creation of new substances.	
		Day 26: Justify that chemical change involves breaking and forming bonds that produce new substances.	
		Describe a valence electron as an electron in the outer shell of an atom that can take part in formation of bonds.	5
		Day 27: Identify valence electrons as electrons found in the outermost shell of an atom.	
		Day 28: Identify the valence shell as the outermost shell of an atom.	
		Day 29: Describe the role of valence electrons in chemical bond formation.	
		Day 30: Explain how valence electrons enable atoms to form chemical bonds.	
		Day 31: Discuss how the number of valence electrons influences the bonding behavior of atoms.	
		DAY 32: (Day 17-31): SUMMATIVE TEST	1

Chemical Bonding: Ionic Compounds and Covalent Compounds	Week 7-10	Identify the number of valence electrons of oxygen based on its position in the periodic table	3
		Day 33: Identify the position of oxygen in the periodic table.	
		Day 34: State the number of valence electrons of oxygen.	
		Day 35: Identify the valence electrons of oxygen based on its group number.	
		Explain the formation of ions as either the loss or gain of electrons to produce ionic bonds, using examples, such as the formation of sodium chloride.	3
		Day 36: Identify ions as charged particles formed when atoms gain or lose electrons	
		Day 37: State that Na ⁺ is a sodium ion with a +1 charge and Cl ⁻ is a chloride ion with a -1 charge and other similar compounds	
		Day 38: Explain the formation of ions as either the loss or gain of electrons to produce ionic bonds, using examples, such as the formation of sodium chloride.	
		Write the chemical formula and chemical names of some common ionic compounds, including sodium chloride (NaCl), magnesium oxide (MgO), potassium chloride (KCl) and magnesium chloride (MgCl₂).	3
		Day 39: Identify the chemical formula and names of some common ionic compounds.	
		Day 40: Explain how the charges of ions determine the chemical formula and name of common ionic compounds.	
		Day 41: Apply the criss-cross method in writing the correct chemical formulas of common ionic compounds.	
		Explain the formation of covalent bonds using a molecule of water and a molecule of carbon dioxide.	3
		Day 42: Define covalent bond as the sharing of electrons between nonmetal atoms.	
		Day 43: Describe the bonding patterns in water and carbon dioxide based on the number of shared electron pairs	
		Day 44: Explain the formation of covalent bonds using a molecule of water and a molecule of carbon dioxide.	
		Write the chemical formula and chemical name of some common covalent compounds, including water (H₂O), carbon dioxide (CO₂), and ammonia (NH₃).	3
		Day 45: Identify the prefix system used in the naming covalent compounds.	

		Day 46: Differentiate between ionic and covalent bonding nomenclature rules to ensure prefixes are applied only to molecular compounds.	
		Day 47: Write the correct chemical formulas and names of common covalent compounds.	
		Show by using models that ionic compounds form crystalline structures whereas covalent compounds form individual molecules.	5
		Day 48: Describe the characteristics of ionic and covalent compounds	
		Day 49 and Day 50: Differentiate ionic and covalent compounds based on structure and bonding	
		Day 51 and Day 52: Show by using models that ionic compounds form crystalline structures whereas covalent compounds form individual molecules.	
Chemical Bonding: Metallic Bonds	Week 11	Explain properties of metals in terms of their structure and metallic bonding (sea of electrons model).	2
		Day 53: Identify the common properties of metals.	
		Day 54: Describe the physical properties of metals such as luster, malleability, ductility and conductivity.	
		Investigate the properties of ionic, covalent, and metallic substances, such as melting point, hardness, electrical and thermal conductivity.	3
		Day 55: List the typical properties (melting point, hardness, electrical and thermal conductivity) of ionic, covalent and metallic substances.	
		Day 56: Explain how bonding type (ionic, covalent, metallic) influences the physical properties of substances.	
		Day 57: Conduct investigations on the melting point, hardness, and electrical and thermal conductivity of ionic, covalent, and metallic substances.	

Reviewed and checked by:

GENOVIE G. TAGUM
Education Program Supervisor- Science